Real-Time ML Analytics Dashboard – Customer Churn System

Repository: [Real-Time-ML-Analytics-Dashboard-Customer-Churn-System]
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Language: Python
Purpose: Real-time customer churn prediction with an interactive dashboard for analytics and visualization.

📌 Project Overview

This project implements a **real-time machine learning system** to predict customer churn. It integrates a backend API with a frontend dashboard, enabling businesses to **monitor churn risk**, identify customers likely to leave, and take proactive measures to retain them. The system provides **live analytics** and visualization of customer churn trends.

Key features:

- Real-time customer churn prediction
- Interactive dashboard for visual analytics
- API-driven backend for data processing and model inference
- Easy integration with existing customer databases

🏗 Project Structure

```
Real-Time-ML-Analytics-Dashboard-Customer-Churn-System/
├── app/              # Backend API & ML logic
│   ├── main.py       # API endpoints
│   ├── models.py     # Data models/classes
│   ├── ml_utils.py   # ML functions (load model, predict)
│   ├── database.py   # Database connection & storage
│   └── schemas.py    # Request/response schemas
├── dashboard/        # Frontend dashboard
│   └── app.py        # UI logic and visualization
```

```
├── .gitignore
└── README.md
```

- **Backend (app/)**: Handles data ingestion, preprocessing, ML model inference, and API responses.
- **Dashboard (dashboard/)**: Displays predictions and analytics in real time using interactive charts and tables.

⚙️ Backend Details

- **main.py** – Entry point for REST API. Handles requests from the dashboard.
- **models.py** – Defines data structures for customers and predictions.
- **ml_utils.py** – Contains functions for loading the trained model, preprocessing input data, and predicting churn probabilities.
- **database.py** – Manages connections to the database and stores customer records.
- **schemas.py** – Ensures proper request validation and response formatting.

The backend processes customer records, predicts churn probability, and returns structured results for dashboard visualization.

📊 Dashboard Details

- **app.py** – Frontend interface for visualizing churn analytics.
- Integrates with backend API to fetch real-time predictions.
- Likely uses **Plotly, Dash, or Streamlit** for interactive charts and dashboards.
- Key features:

    - Visualizing churn risk scores per customer
    - Summary analytics and trends
    - Filtering and segmentation options

🛠 How It Works

1. **Data Input:** Dashboard sends customer data to the backend API.
2. **Data Processing:** Backend preprocesses input for model compatibility.
3. **ML Prediction:** Trained ML model predicts the churn risk score or category.
4. **Visualization:** Dashboard displays predictions and trends in real-time.

This pipeline allows businesses to monitor customer churn continuously and take action proactively.

📌 Recommendations for Improvement

- Add **detailed README** with installation and usage instructions.
- Provide **example dataset** for testing.
- Include **model training scripts** and details on feature engineering.
- Include **dashboard screenshots** for clarity.

⚡ Summary

This project is a **full-stack ML system** combining backend APIs, machine learning, and interactive dashboard visualization. It enables businesses to **analyze customer churn in real-time**, identify risks, and take informed retention actions

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ERRORS:

1.I face the problems in model train with random forest

I solve it with retrain the model.

2. I face the problems in database the data is not save in database,so I create a new database in postgresql.

3.I face the problems in streamlit .